

Markowitz-based portfolio selection with minimum transaction lots, cardinality constraints and regarding sector capitalization using genetic algorithm

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Abstract(1/2)

Heuristic algorithms strengthen researchers to solve more complex and combinatorial problems in a reasonable time. Markowitz's Mean-Variance portfolio selection model is one of those aforesaid problems. Actually, Markowitz's model is a nonlinear(quadratic) programming problem which has been solved by a variety of heuristic and non-heuristic techniques. In this paper a portfolio selection model which is based on Markowitz's portfolio selection problem including three of the most important limitations is considered. The results can lead Markowitz's model to a more practical one.

Abstract(2/2)

Minimum transaction lots, cardinality constraints(both of which have been presented before in other researches) and market(sector) capitalization(which is proposed in this research for the first time as a constraint for Markowitz model), are considered in extended model. No study has ever proposed and solved this expanded model. To solve this mixed-integer nonlinear programming(NP-Hard), a corresponding genetic algorithm(GA) is utilized.

Computational study is performed in two main parts; first verifying and validating proposed GA and second, studying the applicability of presented model using large scale problems.

The proposed model

$$Z = \min \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij}, \quad (1)$$

$$\sum_{i=1}^N x_i c_i r_i \geq bR, \quad (2)$$

$$\sum_{i=1}^N x_i c_i \leq b, \quad (3)$$

$$w_i = \frac{c_i x_i}{\sum_{i=1}^N x_i c_i}, \quad i = (1, 2, \dots, N), \quad (4)$$

$$\sum_{i=1}^N z_i = k, \quad (5)$$

$$B_{\text{lower}_i} - M(1 - z_i) \leq x_i c_i \leq B_{\text{upper}_i} + M(1 - z_i), \quad i = (1, 2, \dots, N), \quad (6)$$

$$\frac{x_i}{M} \leq z_i \leq x_i, \quad i = (1, 2, \dots, N), \quad (7)$$

$$\frac{\sum_{i \in j} z_i}{M} \leq y_j \leq M \sum_{i \in j} z_i, \quad j = (1, 2, \dots, s), \quad (8)$$

$$\sum_{i \in [j]} w_i + (1 - y_j) \geq \sum_{i \in [j+1]} w_i, \quad j = (1, 2, \dots, s-1), \quad (9)$$

minimum
transaction
lots

cardinality
constraints

sector
capitalization

N	be the total number of securities available
r_i	be the expected return of security “ i ” ($i = 1, \dots, N$)
σ_{ij}	be the covariance between securities “ i ” and “ j ” ($i = 1, \dots, N$; $j = 1, \dots, N$)
R	be the minimum desired expected return (of investor)
B	be the total available budget
w_i	be the weight of security “ i ” in portfolio
c_i	be the price of security “ i ” multiply by minimum transaction lots (price of Lot i)
x_i	be the number of minimum transaction lots of Lot “ i ” which model suggest to purchase
B_{upper_i}	be the maximum permitted amount of investing on security i
B_{lower_i}	be the minimum permitted amount of investing on security i
z_i	be a binary variable which will be 1 if any of security “ i ” ($i = 1, \dots, N$) is held and 0 otherwise
M	be a large number
K	be the desired number of securities in the portfolio
S	be the total number of available sectors
y_j	be a binary variable which will be 1 if any security of sector “ j ” is held and 0 otherwise
i	be the index of securities
j	be the index of sectors

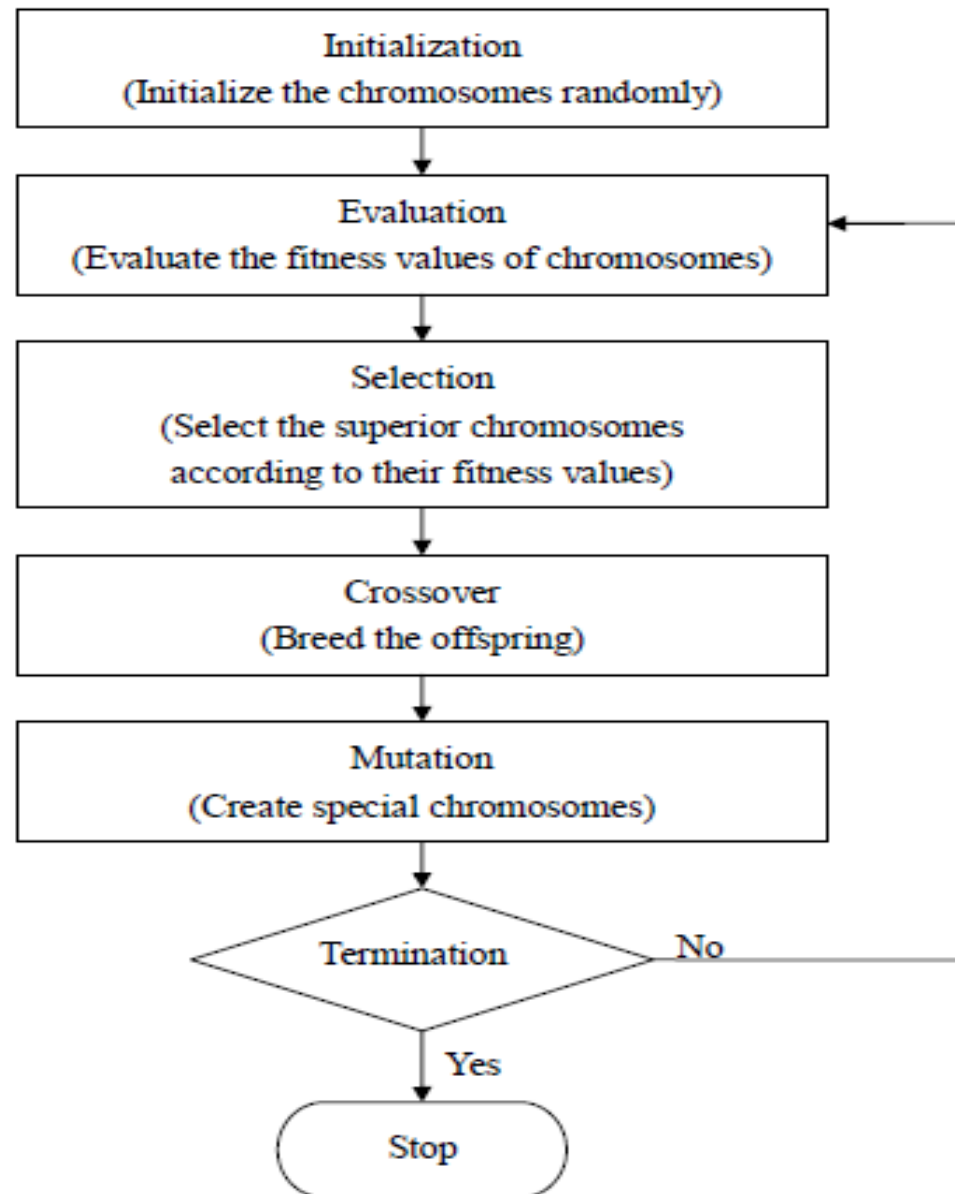
Sector capitalization

- Investors tend to invest in sectors with higher value to reduce their risk of investment.(Chang et al. 2000; Oh et al., 2006)
- The sector with more capitalization should have more proportion in ultimate portfolio if it is selected.

Example(presented in Eq. (9)) :

- Sector 1 has more capitalization than sector 2
- The total weight of securities in sector 1 should more than the total weight in sector 2
- But if there is no securities in portfolio from sector 1, the limitation will be disabled.

Genetic algorithm



Experiment

Parameter setting(use 150 stocks, expect return: 5%)

- GA converged in more than 500 iterations.
- Cross over: 100%
- Mutation : 50%
- Population sizes: acceptable in 20 and 30

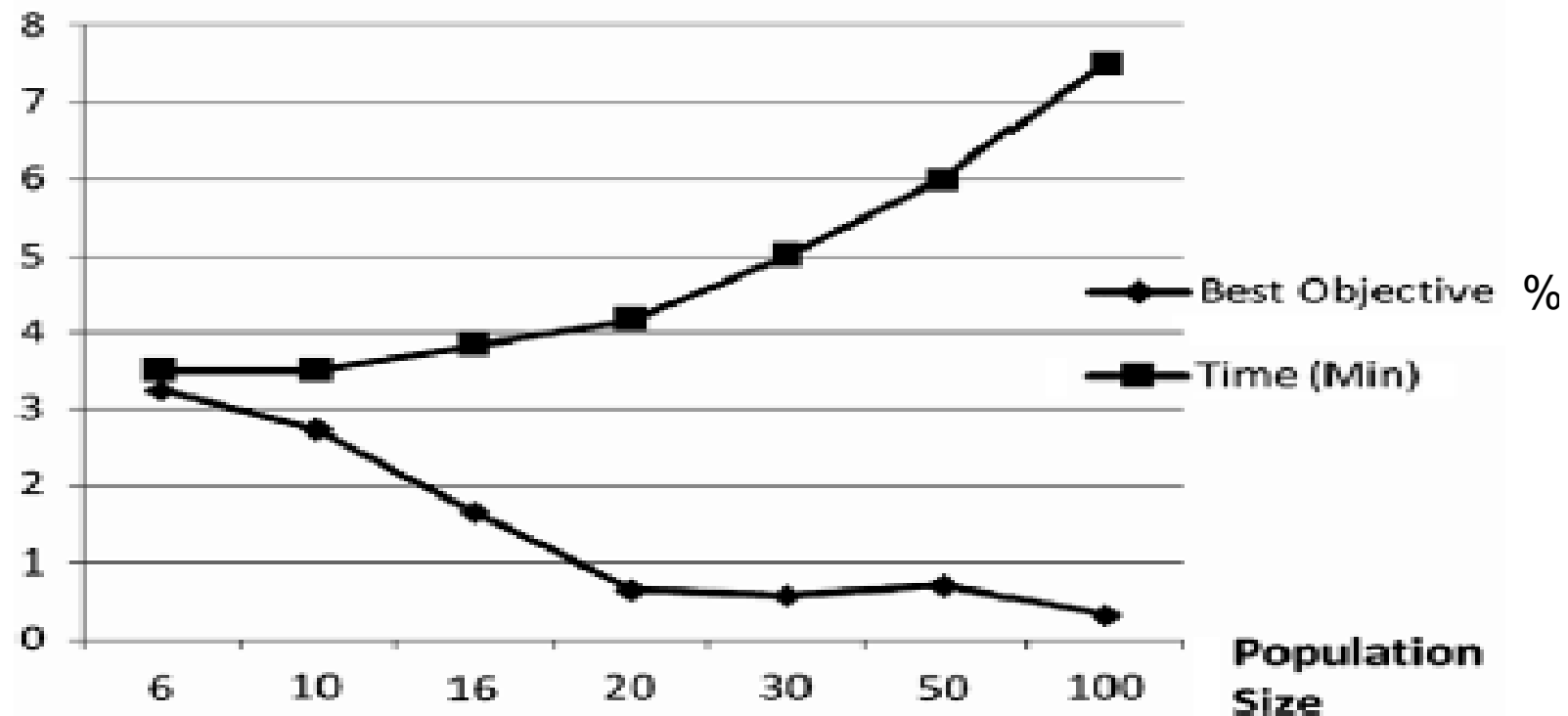


Fig 1. Population size and its effects on best objective and time of solving.

Table 1
Population size and its effects on best objective and time of solving

Population size	6		10		16		20	
	Best objective	Time (min)	Best objective	Time (min)	Best objective	Time (min)	Best objective	Time (min)
$k = 3$	0.0664	3.5	0.0734	3.5	0.02	3.5	0.0149	4
$k = 4$	0.0085	3.5	0.0011	3.5	0.0172	4	0.0032	4
$k = 5$	0.0223	3.5	0.0074	3.5	0.012	4	0.0016	4.5
Average	0.0324	3.5	0.0273	3.5	0.0164	3.83333	0.00657	4.16667

Table 2
Population size and its effects on best objective and time of solving

Population size	30		50		100	
	Best objective	Time (Min)	Best objective	Time (Min)	Best objective	Time (Min)
$k = 3$	0.0137	4.5	0.0078	5	0.0037	6.5
$k = 4$	0.0022	4.5	0.0122	5	0.0025	7
$k = 5$	0.0016	6	0.0013	8	0.0036	9
Average	0.00583	5	0.0071	6	0.00327	7.5

Compare GA and LINGO

- 9 stocks in 3 sectors with sorted sector capitalization is selected.
- The average return and variance of all stocks are **11% and 8.3%**, respectively.
- **LINGO** : a comprehensive tool designed to help you build and solve linear, optimization models quickly, easily, and efficiently
- http://www.lindo.com/index.php?option=com_content&view=article&id=2&Itemid=10

Table 3

The results of experiments in LINGO and the proposed GA for $k = 2$

R	LINGO	GA	Difference
5	0.0000001688	0.0000074850	0.0000073162
6	0.0000001961	0.0046000000	0.0045998039
7	0.0000000031	0.0003586800	0.0003586769
8	0.0000000650	0.0117000000	0.0116999350
9	0.0000008759	0.0000318530	0.0000309771
10	0.0000000002	0.0117000000	0.0116999998
11	0.0000000089	0.0055000000	0.0054999911
12	0.0000003849	0.0021000000	0.0020996151
Average			0.0044995394

- **GA** :

Iteration: 1000

Population size: 30

Cross over: 100%

Mutation : 50%

Table 4

The results of experiments in LINGO and the proposed GA for $k = 3$

R	LINGO	GA	Difference
5	0,0068832220	0,0070000000	0,0001167780
6	0,0068832210	0,0155000000	0,0086167790
7	0,0068833860	0,0085000000	0,0016166140
8	0,0068833860	0,0094000000	0,0025166140
9	0,0068833860	0,0132000000	0,0063166140
10	0,0050166170	0,0082000000	0,0031833830
11	0,0050166170	0,0069000000	0,0018833830
12	0,0068832220	0,0341000000	0,0272167780
Average			0,0064333679

Table 5

The results of experiments in LINGO and the proposed GA for $k = 5$

R	LINGO	GA	Difference
5	0,0369502500	0,0971000000	0,0601497500
6	0,0369502500	0,2780000000	0,2410497500
7	0,0369502500	0,1340000000	0,0970497500
8	0,0369502500	0,1729000000	0,1359497500
9	0,0369502500	0,1085000000	0,0715497500
10	0,0369502500	0,0812000000	0,0442497500
11	0,0369502500	0,0812000000	0,04424975
12	0,0870655800	0,2590800000	0,17201442
Average			0,0777453338

Applicability in large scale problems

- 500 and 2000 stocks(N) with 10 sectors are selected
- Average return and variance of market are around 7% and 15%

Table 6

The applicability of the proposed GA of the model in real markets

Number of stocks	Iteration	k	R	Time (min)	Objective
500	1000	3	5	6	0.0157
500	1000	3	7	7	0.0025
500	1000	3	9	7	0.017
500	1000	5	5	6	0.0216
500	1000	5	7	7	0.0069
500	1000	5	9	8	0.0201
2000	500	3	5	16	0.1136
2000	500	3	7	16	0.0877
2000	500	3	9	16	0.0241
2000	500	5	5	16	0.0628
2000	500	5	7	17	0.0405
2000	500	5	9	17	0.0161
Average	-	-	-	11.583	0.035717

Thank You!!